

V3 moneyness performance

Same 6 models · 3 companies · 4 windows · 1.5-year monthly study · classified by call moneyness $m = S/K$

What this report is

- Companion to V3_1p5y_monthly_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets ($n = 0$) show em dashes and have no BEST.

Unchanged experimental design

- Same six models, three companies (SPY, AAPL, MSFT), 4 windows, 1.5-year lookback, monthly recalibration, V3 filters, Euler-corrected Heston step, LSM $n_{\text{paths}} = 10000$ / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7-60, $|S/K - 1| \leq 10\%$). No model-specific re-sampling.
- Pooled $n = 601$ contracts per model (identical keys across models).

Moneyness rule (calls, $m = S / K$)

- DOTM: $m < 0.98$. OTM: $0.98 \leq m < 0.995$. ATM: $0.995 \leq m \leq 1.005$. ITM: $1.005 < m \leq 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \leq 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008-2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$. Ranking uses this only.
- MAE and Bias \$ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model $\times$ Moneyness RMSE% heatmap.
- Analysis 2 — Window $\times$ moneyness: keep the 4 windows separate, pool the three companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2008-2009 Crisis	8	30	72	34	6	150
2013-2014 Normal	0	5	142	3	0	150
2018-2019 Late-cycle	0	4	140	7	0	151
2019-2020 COVID	0	6	139	5	0	150
All regimes	8	45	493	49	6	601

Moneyness rule · American calls, $m = S/K$

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	8	1.3%
OTM	$0.98 \leq m < 0.995$	45	7.5%
ATM	$0.995 \leq m \leq 1.005$	493	82.0%
ITM	$1.005 < m \leq 1.02$	49	8.2%
DITM	$m > 1.02$	6	1.0%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

Table M1 · DOTM · $m < 0.98$ · $n = 8$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	8	38.99	1.0687	+0.1518	+18.27	17.4%	5
GARCH	8	38.88	1.0352	+1.0352	+32.13	16.4%	4
Heston	8	27.37	0.8354	+0.8354	+21.90	20.4%	2
Merton	8	35.89	0.9641	+0.0252	+15.01	15.3%	3
GARCH-Merton	8	49.69	1.4663	+1.4663	+43.21	13.6%	6
Heston-Merton	8	26.73	0.8325	+0.8325	+21.13	16.0%	1

Table M2 · OTM · $0.98 \leq m < 0.995$ · $n = 45$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	45	65.05	1.7303	+0.8262	+34.72	21.6%	4
GARCH	45	72.45	1.9400	+1.6133	+49.67	24.7%	5
Heston	45	37.34	1.1727	+0.7013	+23.09	20.2%	2
Merton	45	63.01	1.7017	+0.7294	+32.09	16.9%	3
GARCH-Merton	45	82.67	2.1269	+1.8807	+62.00	23.1%	6
Heston-Merton	45	36.89	1.1665	+0.5824	+20.12	15.4%	1

Table M3 · ATM · $0.995 \leq m \leq 1.005$ · $n = 493$ · BEST = Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	493	49.51	2.0094	+1.2989	+32.17	24.0%	4
GARCH	493	57.86	2.1491	+1.8497	+39.76	23.4%	5
Heston	493	47.58	1.9916	+1.6436	+34.64	23.6%	2
Merton	493	46.46	1.8715	+1.1231	+28.85	19.5%	1
GARCH-Merton	493	128.41	3.0863	+2.8985	+56.65	22.0%	6
Heston-Merton	493	49.36	1.8674	+1.4954	+32.85	17.5%	3

Bias \$ = mean(model – market). Bias% = 100 × mean((model – market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4-M5

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

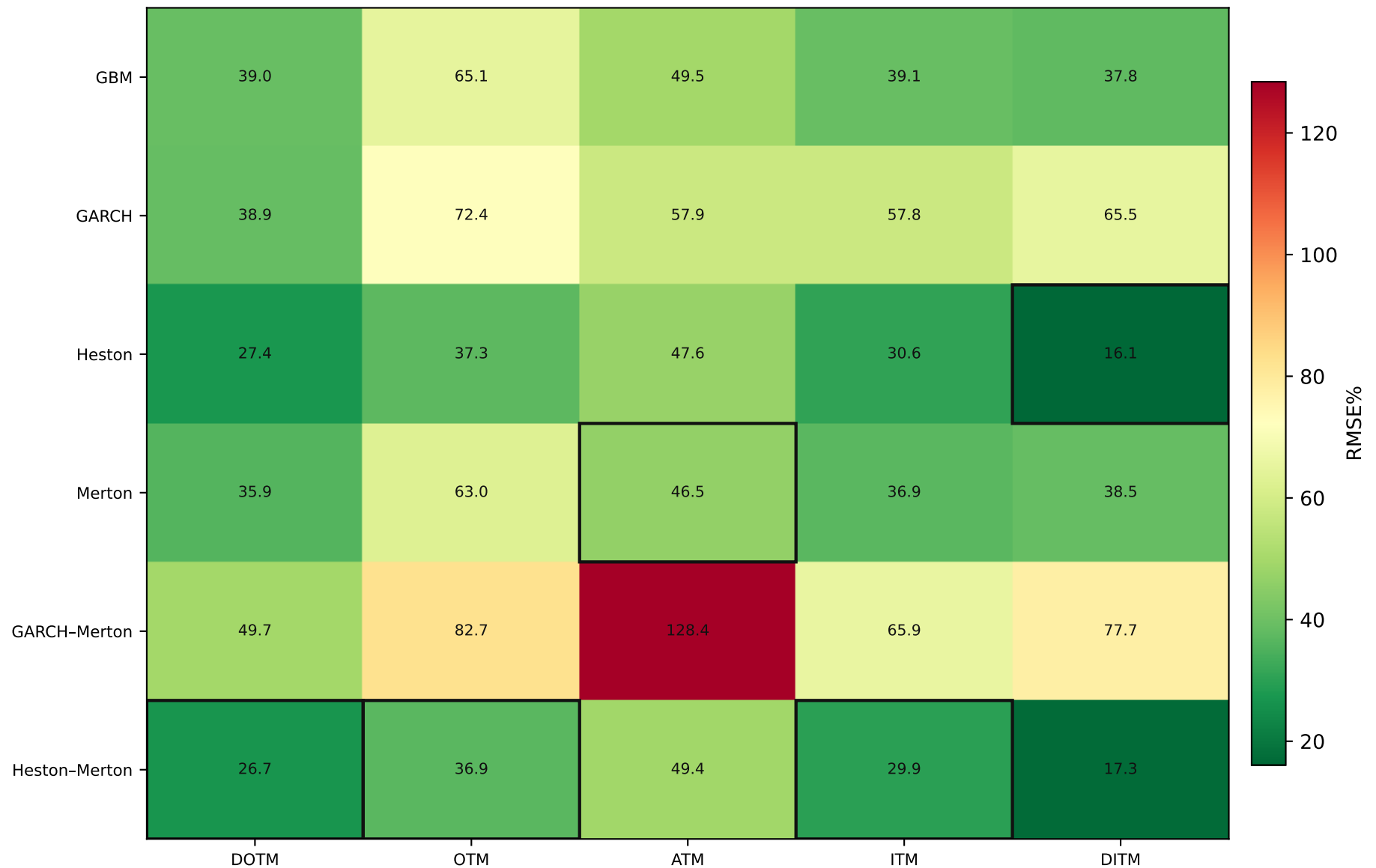
Table M4 · ITM · $1.005 < m \leq 1.02$ · $n = 49$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	49	39.08	1.3788	+0.6083	+16.37	26.8%	4
GARCH	49	57.82	1.6448	+1.5143	+44.04	29.2%	5
Heston	49	30.59	1.2225	+0.7848	+15.66	26.6%	2
Merton	49	36.85	1.3139	+0.4977	+13.73	20.8%	3
GARCH-Merton	49	65.95	1.9885	+1.9081	+54.61	26.5%	6
Heston-Merton	49	29.95	1.1187	+0.6541	+14.47	19.4%	1

Table M5 · DITM · $m > 1.02$ · $n = 6$ · BEST = Heston

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	6	37.81	0.8669	-0.2089	+16.37	28.7%	3
GARCH	6	65.46	1.9584	+1.9584	+55.63	30.4%	5
Heston	6	16.09	0.3200	+0.2097	+8.73	27.5%	1
Merton	6	38.51	0.9122	-0.3251	+14.50	23.0%	4
GARCH-Merton	6	77.72	2.2914	+2.2914	+67.88	22.4%	6
Heston-Merton	6	17.29	0.3896	+0.1808	+8.83	20.1%	2

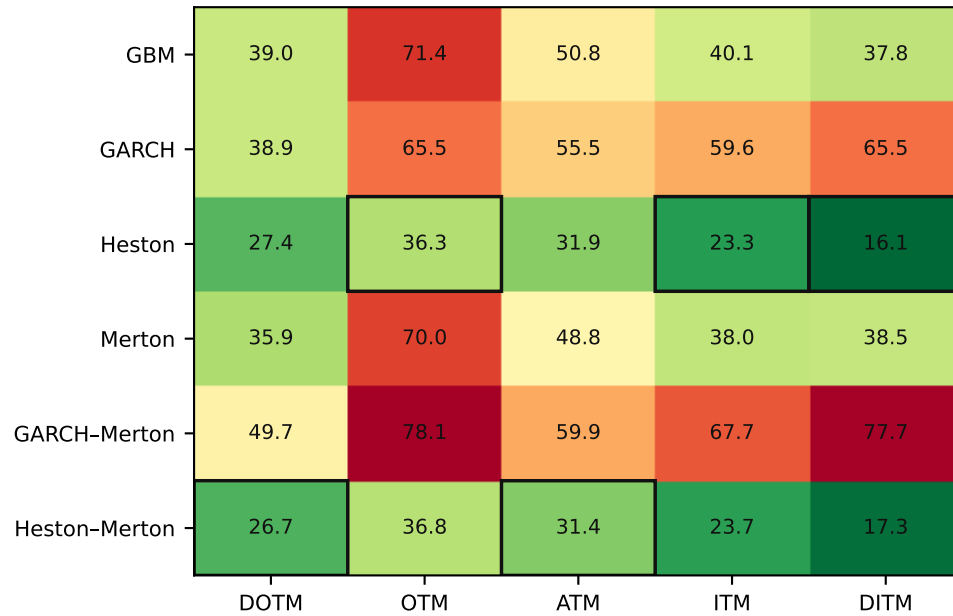
ITM/DITM: $S > K$. DOTM/OTM: $S < K$. ATM: $|S/K - 1| \leq 0.5\%$.



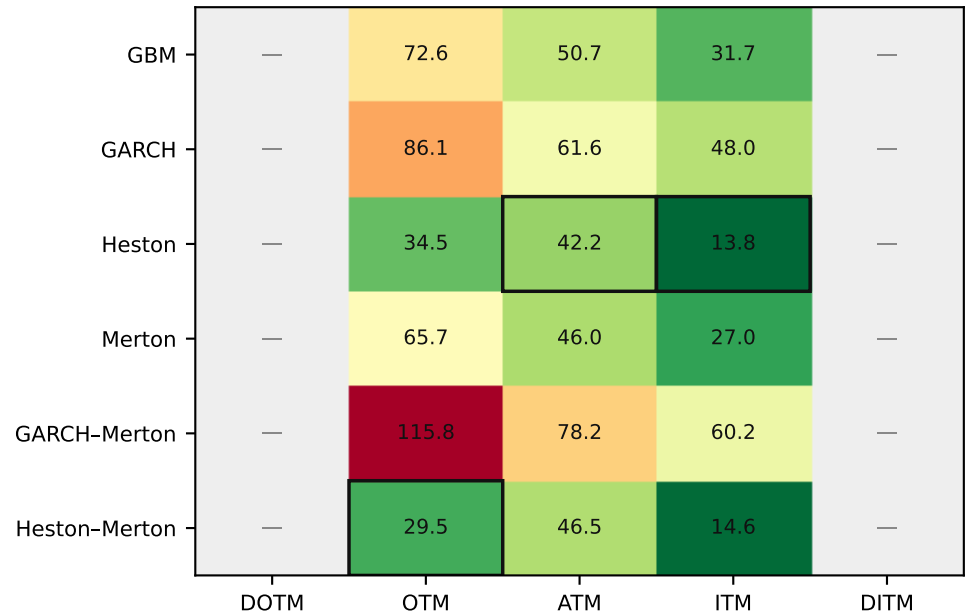
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)

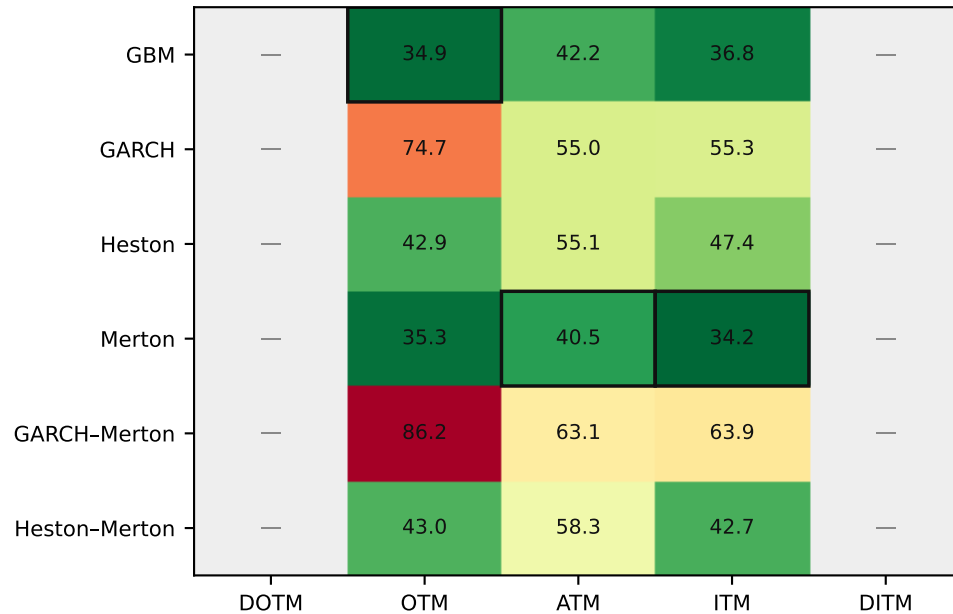
2008-2009 · Crisis



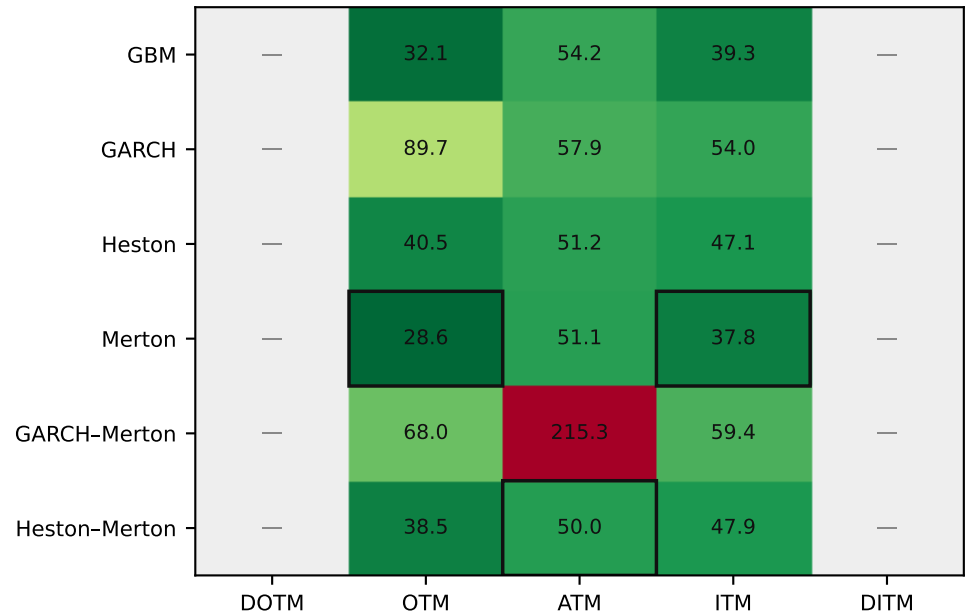
2013-2014 · Normal



2018-2019 · Late-cycle



2019-2020 · COVID



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Analysis 2 · Does the ranking change with moneyness inside each window?

Table M6 · Best model (lowest RMSE%) by window × moneyness

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	Heston-Merton	Heston	Heston-Merton	Heston	Heston
2013-2014 Normal	—	Heston-Merton	Heston	Heston	—
2018-2019 Late-cycle	—	GBM	Merton	Merton	—
2019-2020 COVID	—	Merton	Heston-Merton	Merton	—

Table M7 · n contracts by window × moneyness (shared across models)

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	8	30	72	34	6
2013-2014 Normal	0	5	142	3	0
2018-2019 Late-cycle	0	4	140	7	0
2019-2020 COVID	0	6	139	5	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

Does performance change with moneyness?

- DOTM (n = 8): BEST = Heston-Merton with RMSE% = 26.73, Bias% = +21.13.
- OTM (n = 45): BEST = Heston-Merton with RMSE% = 36.89, Bias% = +20.12.
- ATM (n = 493): BEST = Merton with RMSE% = 46.46, Bias% = +28.85.
- ITM (n = 49): BEST = Heston-Merton with RMSE% = 29.95, Bias% = +14.47.
- DITM (n = 6): BEST = Heston with RMSE% = 16.09, Bias% = +8.73.
- Overall, the ranking does change with moneyness: DOTM → Heston-Merton, OTM → Heston-Merton, ATM → Merton, ITM → Heston-Merton, DITM → Heston.
- Inside regimes, the BEST model also changes with moneyness in: 2008-2009, 2013-2014, 2018-2019, 2019-2020.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008-2009 quotes and should be read as descriptive, not as a high-powered test.